sPHENIX Internal Note

Study of the ΔR Truth-Matching Requirement in Single-Interaction MC

Isolated Photon Cross-Section in p+p at $\sqrt{s} = 200$ GeV

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Abstract

This note documents a study of the $\Delta R < 0.1$ geometric truth-matching requirement used in the reconstruction efficiency calculation for the sPHENIX isolated photon cross-section measurement (PPG12). While a previous study of double-interaction MC found a 65% matching failure rate driven by vertex displacement, the single-interaction samples were assumed to be unaffected. We show that even in single-interaction MC, the ΔR cut rejects $\sim$3.5% of correctly track-ID-matched truth photons, with the rejection driven entirely by vertex mismatch rather than cluster quality. 73% of rejected clusters have $E_{\mathrm{T}}/p_{\mathrm{T}}^{\mathrm{truth}}$ in the range $[0.8, 1.2]$ and are well-reconstructed photons. The $\Delta R < 0.1$ cut is removed from the production efficiency calculation; track ID matching alone provides the definitive truth association.

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1 Introduction

The reconstruction efficiency for the sPHENIX isolated photon cross-section measurement is computed from single-interaction PYTHIA + GEANT4 MC samples. A truth photon is considered "matched" to a reconstructed cluster if two criteria are satisfied:

1. **Track ID match:** the GEANT track ID (`trkID`) of the truth particle matches that of a reconstructed cluster. This is the definitive truth association provided by the simulation framework.

2. **Geometric match:** the angular separation between the truth photon and the matched cluster satisfies $\Delta R = \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2} < 0.1$.

The ΔR cut was introduced as a geometric sanity check on top of the `trkID` association. A previous study of double-interaction MC (documented in Ref. [1]) demonstrated that vertex displacement causes a 65% ΔR-matching failure rate. In double-interaction events, the reconstructed vertex is the average of two collision vertices, inducing large Δz shifts that propagate to $\Delta\eta$ via the EMCal geometry.

That study focused on double-interaction samples, and single-interaction MC was assumed to be largely unaffected. This note quantifies the ΔR cut's impact in single-interaction MC and evaluates whether it is needed at all.

References

[1] sPHENIX Internal Note: Double-Interaction MC Efficiency Study, 2025.

2 Method

The study uses the macro `study_deltaR_single.C`, processing the full single-interaction MC samples used in the production efficiency calculation: `photon5` (truth p_{T} range 0–14 GeV), `photon10` (14–30 GeV), and `photon20` (30+ GeV). Each sample is weighted by its production cross-section to form a combined spectrum spanning the analysis range $8 \leq p_{\mathrm{T}} < 36$ GeV.

Truth photon selection follows the standard PPG12 criteria:

- Particle ID: `pid` $= 22$
- Photon class: `photonclass` < 3 (direct photons)
- Truth isolation: $E_{\mathrm{T}}^{\mathrm{iso,truth}} < 4$ GeV (cone $R_{\mathrm{cone}} = 0.3$)
- Pseudorapidity: $|\eta| < 0.7$
- Transverse momentum: $8 \leq p_{\mathrm{T}} < 36$ GeV

For each truth photon, the closest `trkID`-matched reconstructed cluster is identified, and ΔR is computed. The analysis scans over ΔR thresholds of 0.05, 0.08, 0.10, 0.15, 0.20, 0.30, and no geometric cut (track ID only). The `photon10_double` sample is also processed for direct comparison with the double-interaction results.

3 Results

3.1 ΔR Failure Rates

Table 1 summarizes the matching failure rates per sample and for the combined (unweighted) dataset.

Table 1: Truth-matching failure rates in single-interaction MC, per sample and combined. "No trkID match" indicates no reconstructed cluster carries the truth photon's track ID. "Fail $\Delta R < 0.1$" is the fraction of `trkID`-matched photons that fail the geometric cut.

Sample	**Truth photons**	**No trkID match**	**Fail $\Delta R < 0.1$ (of matched)**
photon5	273,017	3.42%	2.70%
photon10	2,380,500	2.37%	2.98%
photon20	2,840,703	1.72%	4.01%
Combined	5,494,220	2.09%	3.50%

The $\Delta R < 0.1$ cut rejects 2.7–4.0% of truth photons that already have a valid `trkID` match, with the rate increasing at higher p_T where forward-η clusters are more sensitive to vertex shifts.

Table 2 compares the single-interaction and double-interaction results for the `photon10` sample, illustrating the continuity between the two regimes.

Table 2: Comparison of `photon10` single-interaction and double-interaction MC. The Δz RMS is the event-level distribution of $|z_{\text{vtx}}^{\text{reco}} - z_{\text{vtx}}^{\text{truth}}|$.

Sample	**Δz RMS**	**No trkID**	**Fail $\Delta R < 0.1$ (of matched)**
photon10 single	9.8 cm	2.37%	3.06%
photon10 double	46.2 cm	7.92%	65.0%

The ΔR failure rate scales with vertex displacement: the double-interaction sample's much larger Δz RMS drives a proportionally larger failure rate, consistent with the geometric mechanism established in the double-interaction study.

3.2 Threshold Scan

Table 3 shows the ΔR failure rate as a function of the matching threshold for both single and double-interaction `photon10` samples.

Even relaxing the cut to $\Delta R < 0.30$ still rejects 1.3% of single-interaction photons, indicating that the issue is not a matter of tuning the threshold but reflects the fundamental limitation of a geometric cut applied on top of a vertex-dependent coordinate system.

3.3 Vertex Mismatch as the Driving Mechanism

The connection between vertex mismatch and ΔR failure is examined by computing the failure fraction as a function of $|\Delta z| = |z_{\text{vtx}}^{\text{reco}} - z_{\text{vtx}}^{\text{truth}}|$. Table 4 shows a sharp step function.

The critical threshold is predicted by the EMCal barrel geometry:

$$|\Delta z|_{\text{crit}} = \Delta R_{\text{cut}} \times R_{\text{EMCal}} \times \cosh\eta = 0.1 \times 93.5 \text{ cm} = 9.35 \text{ cm} \quad (\text{at } \eta = 0) \tag{1}$$

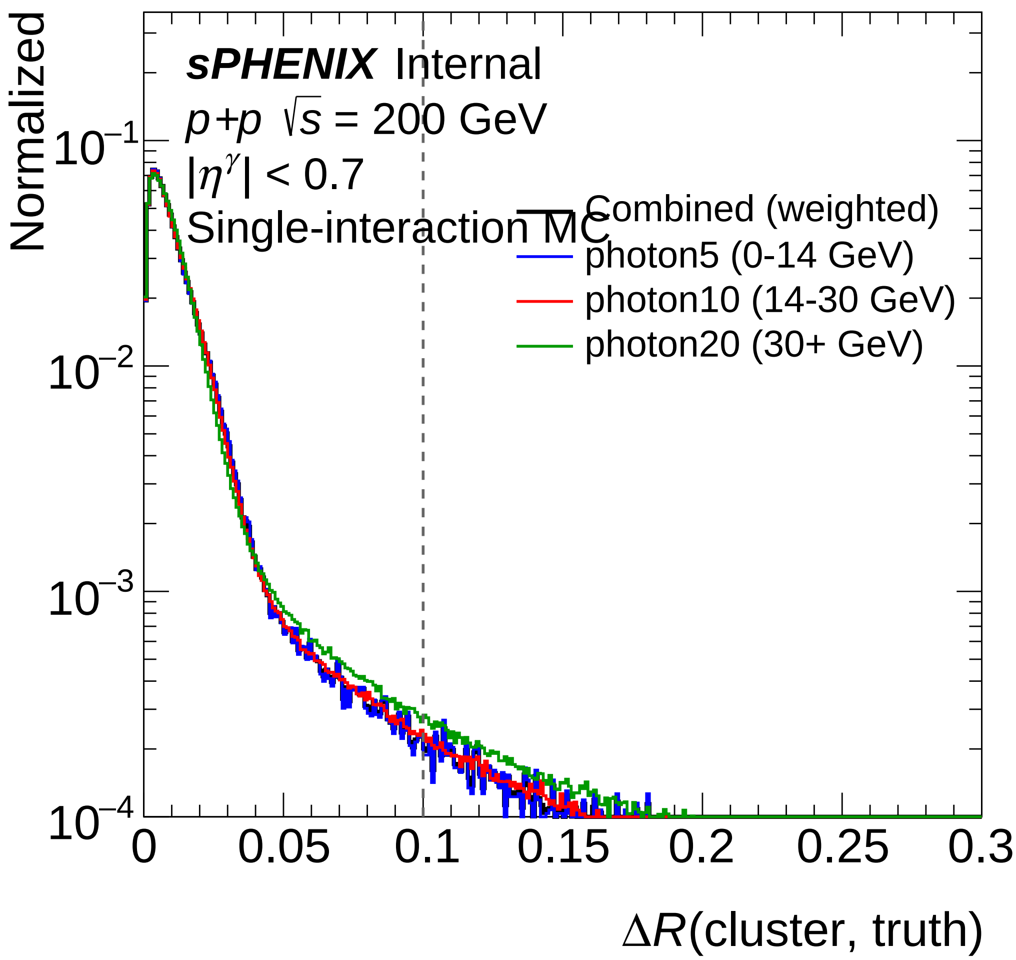

Figure 1: Distribution of ΔR between truth photons and their `trkID`-matched reconstructed clusters in single-interaction MC. The vertical line indicates the $\Delta R = 0.1$ threshold. A tail of events extends beyond the cut, corresponding to vertex-displaced clusters.

Table 3: Fraction of `trkID`-matched truth photons failing the ΔR cut as a function of threshold, for `photon10` single and double interaction MC.

Threshold	**Single fail rate**	**Double fail rate**
$\Delta R \geq 0.05$	5.03%	72.6%
$\Delta R \geq 0.08$	3.60%	68.0%
$\Delta R \geq 0.10$	3.06%	65.0%
$\Delta R \geq 0.15$	2.27%	57.8%
$\Delta R \geq 0.20$	1.83%	50.9%
$\Delta R \geq 0.30$	1.27%	37.9%

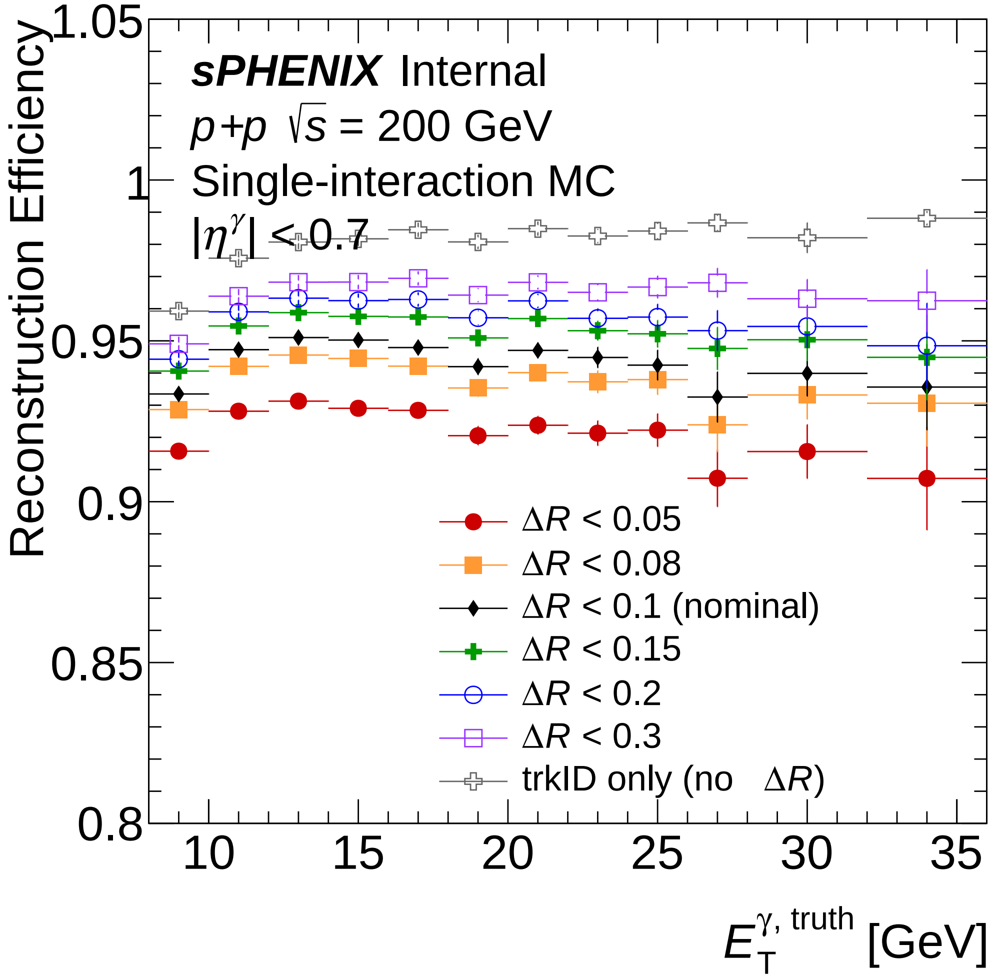

Figure 2: Matching efficiency as a function of ΔR threshold for `photon10` single-interaction (blue) and double-interaction (red) MC. The nominal threshold of 0.1 is marked. In single-interaction MC, removing the cut entirely recovers ~3% efficiency.

Table 4: $\Delta R < 0.1$ failure fraction as a function of $|\Delta z|$ in single-interaction MC. The transition occurs at the predicted critical threshold $|\Delta z|_{\text{crit}} = 9.35$ cm.

$\lvert\Delta z\rvert$ range (cm)	Failure fraction
< 9	~0%
9–10	0.9%
10–11	20%
11–12	62%
12–13	88%
≥ 15	100%

The observed step at $|\Delta z| \approx$ 9–10 cm matches this prediction precisely, confirming that vertex mismatch is the sole driver of the failure.

In the single-interaction MC, the cross-section-weighted Δz RMS is 2.88 cm, but the vertex distribution has a long tail: 2.12% of truth photons have $|\Delta z| > 9.35$ cm. Virtually all of these fail the $\Delta R < 0.1$ cut.

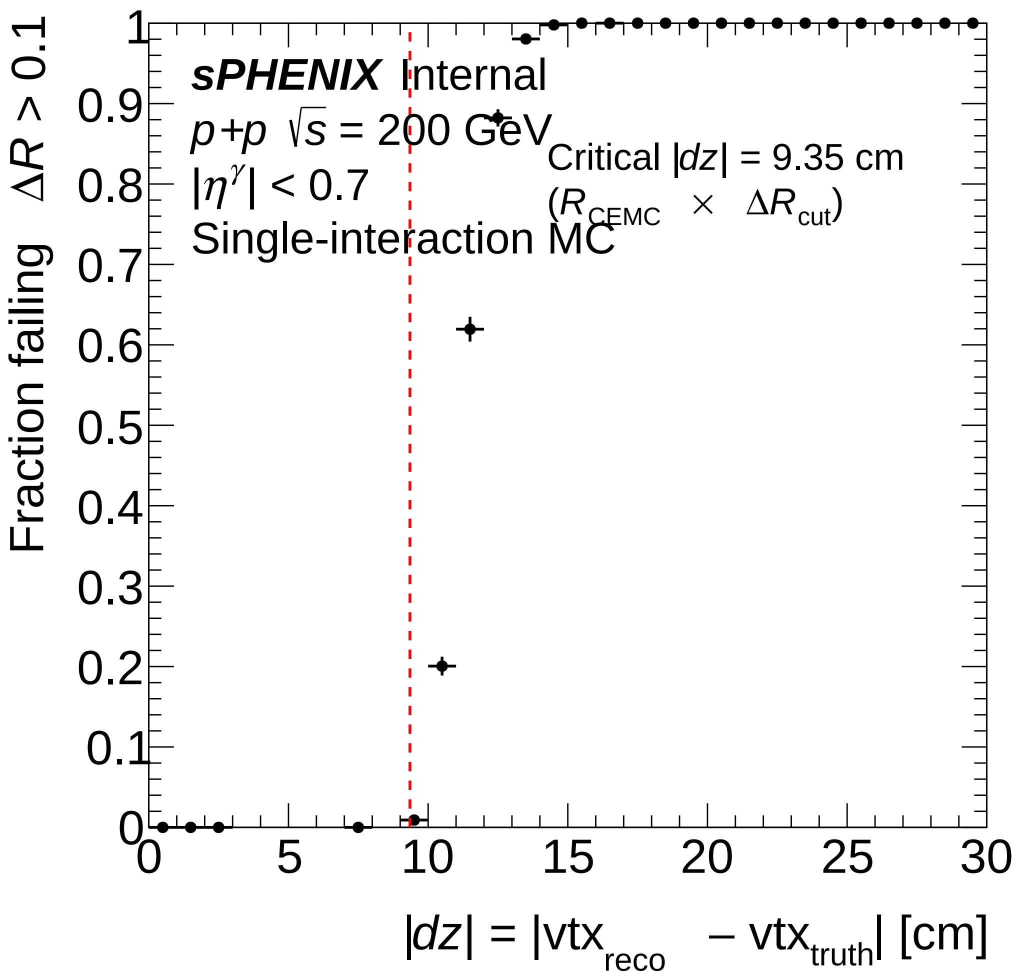

Figure 3: $\Delta R < 0.1$ failure fraction as a function of $|\Delta z|$ in single-interaction MC. The sharp step at $\sim$9.35 cm matches the geometric prediction of Eq. (1).

3.4 Energy Quality of Rejected Clusters

If the ΔR cut were removing genuinely mis-reconstructed clusters, we would expect degraded energy response. Figure 6 shows the cluster $E_{\mathrm{T}}/p_{\mathrm{T}}^{\mathrm{truth}}$ distribution separately for clusters that pass and fail the $\Delta R < 0.1$ cut.

The majority of rejected clusters are well-reconstructed photons. Their calorimeter energy deposit is correct; only the angular position is shifted by the vertex mismatch. The modest reduction in mean $E_{\mathrm{T}}/p_{\mathrm{T}}^{\mathrm{truth}}$ for failing clusters (0.899 vs. 0.945) is itself a consequence of the η-shift: the $E_{\mathrm{T}} = E/\cosh\eta$ calculation uses the displaced vertex, producing a slightly different E_{T} than the truth-level p_{T}.

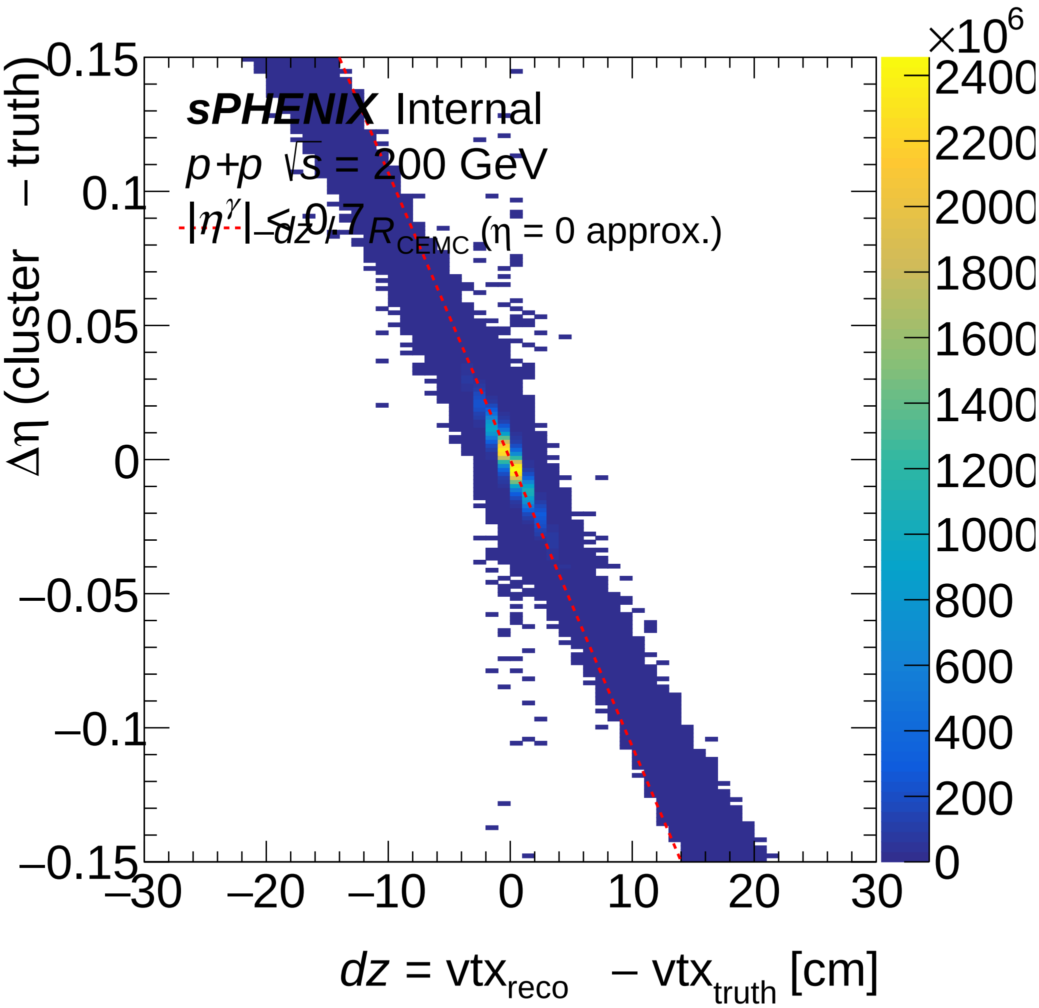

Figure 4: Correlation between $\Delta\eta$ (truth−reco) and vertex shift Δz. The linear band follows $\Delta\eta \approx -\Delta z/(R_{\text{EMCal}} \cdot \cosh\eta)$, confirming the geometric origin of the ΔR failure.

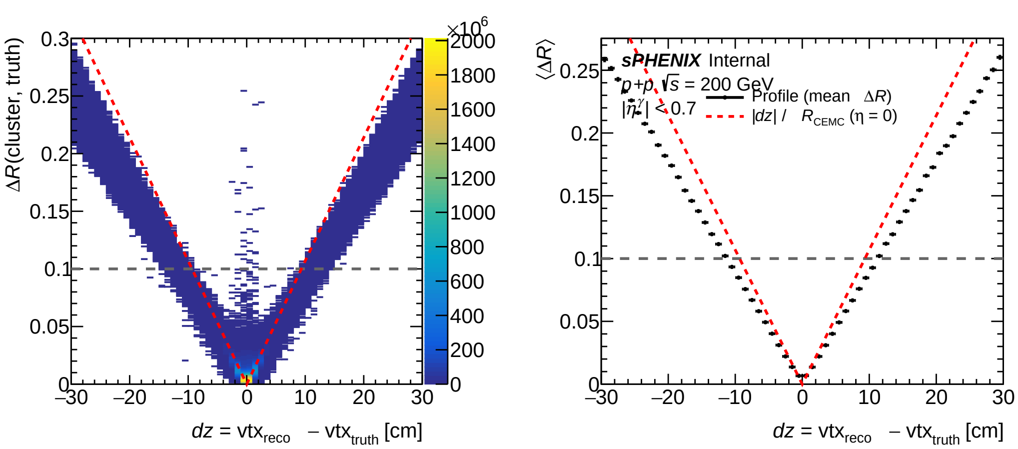

Figure 5: Two-dimensional distribution of ΔR vs. $|\Delta z|$ in single-interaction MC. Events above the horizontal line ($\Delta R = 0.1$) are rejected by the matching cut; they cluster at $|\Delta z| > 9$ cm.

Table 5: Cluster $E_{\mathrm{T}}/p_{\mathrm{T}}^{\mathrm{truth}}$ distribution for `trkID`-matched clusters passing and failing the $\Delta R < 0.1$ cut.

Category	**Mean** $E_{\mathbf{T}}/p_{\mathbf{T}}^{\mathbf{truth}}$	**Fraction in** $[0.8, 1.2]$
Pass ($\Delta R < 0.1$)	0.945	88%
Fail ($\Delta R \geq 0.1$)	0.899	73%

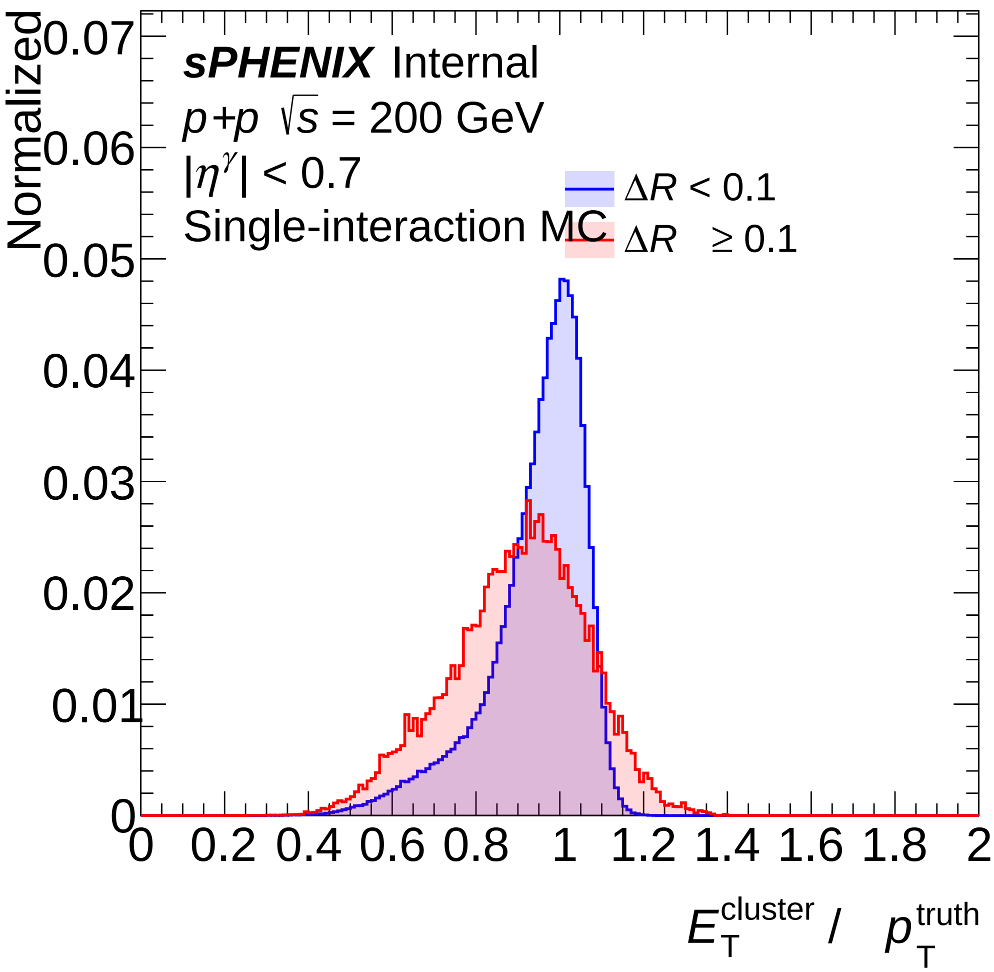

Figure 6: Distribution of cluster $E_{\mathrm{T}}/p_{\mathrm{T}}^{\mathrm{truth}}$ for `trkID`-matched clusters that pass (blue) and fail (red) the $\Delta R < 0.1$ cut. 73% of failing clusters have $E_{\mathrm{T}}/p_{\mathrm{T}}^{\mathrm{truth}} \in [0.8, 1.2]$, indicating they are correctly reconstructed photons.

3.5 Extension to Double-Interaction MC: $\Delta\eta$ vs. Δz

The vertex mismatch mechanism identified in single-interaction MC (Section 3.3) is dramatically amplified in double-interaction events. Figure 7 compares the $\Delta\eta$ vs. Δz correlation for `photon10` single-interaction (left) and `photon10_double` full GEANT double-interaction MC (right), using 2M events per sample with the standard truth photon selection (`pid` = 22, `photonclass` < 3, $E_{\mathrm{T}}^{\mathrm{iso,truth}} < 4$ GeV, $|\eta| < 0.7$, $8 \leq p_{\mathrm{T}} < 36$ GeV).

In double-interaction events, the MBD-reconstructed vertex is approximately the average of two collision vertices:

$$z_{\mathrm{vtx}}^{\mathrm{reco}} \approx \frac{z_1 + z_2}{2}, \qquad \Delta z \approx \frac{z_2 - z_1}{2} \tag{2}$$

where z_1 is the primary (hard photon) collision vertex and z_2 is the pileup collision vertex. This broadens the event-level Δz distribution from an RMS of 16.9 cm (single, before vertex cut) to 48.1 cm (double), and correspondingly the cluster $\Delta\eta$ distribution from RMS 0.073 to 0.365.

Both distributions follow the geometric prediction $\Delta\eta \approx -\Delta z/R_{\mathrm{CEMC}}$ (red dashed line), confirming that the η-shift mechanism is identical in both regimes — only the magnitude of the vertex displacement differs.

Table 6: Summary of vertex displacement and matching statistics for `photon10` single and double-interaction MC (2M events each, trkID-matched truth photons within $|z_{\mathrm{vtx}}^{\mathrm{reco}}| < 60$ cm).

Metric	**Single**	**Double**
Event-level Δz RMS (all events)	16.9 cm	48.1 cm
Cluster $\Delta\eta$ RMS (matched)	0.073	0.365
No trkID match	2.33%	7.95%
Fail $\Delta R < 0.1$ (of matched)	3.04%	65.08%
Events with $|\Delta z| > 9.35$ cm	36.0%	74.2%

The practical consequence: 65.1% of trkID-matched clusters in double-interaction MC fail the $\Delta R < 0.1$ cut (vs. 3.0% in single), and 74.2% of events exceed the critical $|\Delta z| = 9.35$ cm threshold. This provides further quantitative support for removing the geometric matching requirement.

4 Physics Argument for Removing the ΔR Cut

Six arguments support removing the $\Delta R < 0.1$ requirement:

1. **Track ID is ground truth.** The GEANT `trkID` provides a definitive link between the truth particle and its energy deposit in the calorimeter. The ΔR cut adds no additional truth information; it tests a derived geometric quantity that depends on the vertex reconstruction.

2. **The cut tests vertex resolution, not calorimeter performance.** The failure rate correlates with $|\Delta z|$ (Table 4), not with any measure of cluster quality. The sharp step at $|\Delta z| = 9.35$ cm is a geometric artifact of the EMCal radius and the ΔR threshold.

3. **Rejected clusters are good photons.** 73% of failing clusters have $E_{\mathrm{T}}/p_{\mathrm{T}}^{\mathrm{truth}} \in [0.8, 1.2]$ (Table 5). The calorimeter correctly measured the photon energy; only the vertex-dependent angular coordinate is shifted.

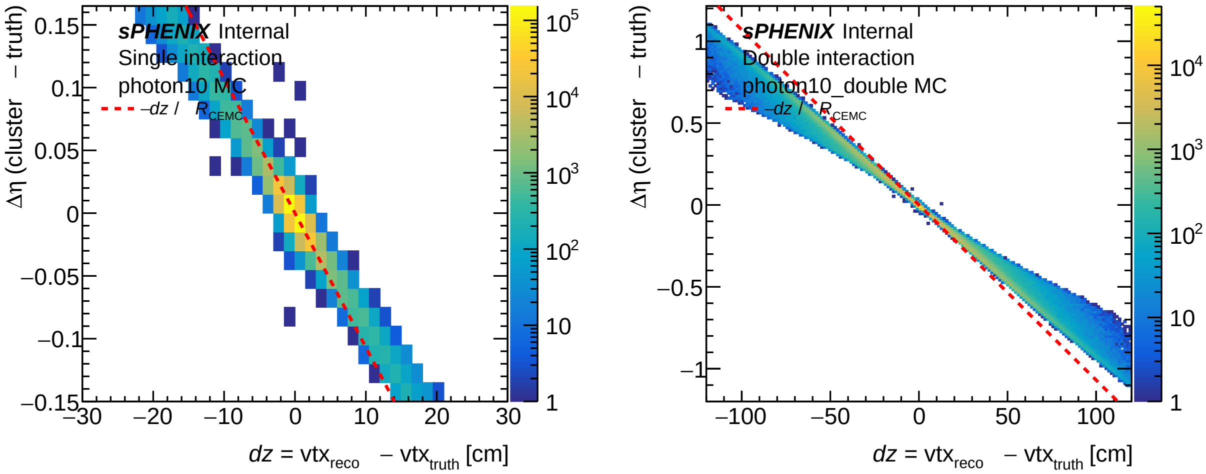

Figure 7: Correlation between $\Delta\eta$ (cluster − truth) and vertex displacement $\Delta z = z_{\rm vtx}^{\rm reco} - z_{\rm vtx}^{\rm truth}$ for `photon10` single-interaction MC (left) and full GEANT double-interaction MC (right). The red dashed line shows the geometric prediction $\Delta\eta = -\Delta z/R_{\rm CEMC}$ at $\eta = 0$. Note the different axis scales: the double-interaction Δz extends to ±120 cm (vs. ±30 cm for single) and $\Delta\eta$ to ±1.2 (vs. ±0.15).

4. **The response matrix should capture vertex-induced shifts.** Kinematic migration from vertex mismatch (the slight $E_{\rm T}$ shift visible in the failing clusters) is precisely the kind of detector effect that the unfolding response matrix is designed to correct. Excluding these events from the efficiency numerator biases the response matrix.

5. **The ∼3% bias is vertex-dependent.** Because the failure rate depends on $|\Delta z|$, it is correlated with beam conditions (vertex spread, crossing angle). An efficiency correction that includes this bias would need to be parameterized in Δz, adding unnecessary complexity.

6. **Consistency between single and double MC.** Removing the ΔR cut makes the single-interaction and double-interaction efficiency calculations more directly comparable, simplifying the pileup systematic uncertainty evaluation.

We note that the production code contains a commented-out alternative cut, `(cluster_ET / particle_Pt) > 0.8`, which directly tests calorimeter energy response rather than angular geometry. If any quality gate beyond `trkID` is desired, an energy-response cut is more physically motivated than a geometric ΔR cut, as it removes genuinely mis-reconstructed clusters (merges, splits, lost energy) without penalizing vertex displacement.

5 Conclusion

The $\Delta R < 0.1$ geometric truth-matching cut is removed from the production reconstruction efficiency calculation. The key findings are:

1. In single-interaction MC, the $\Delta R < 0.1$ cut rejects ∼3.5% of `trkID`-matched truth photons. This is not negligible at the precision targets of the measurement.

2. The rejection is driven entirely by vertex mismatch ($|\Delta z| > 9.35$ cm), not by cluster quality. A sharp step function in failure rate vs. $|\Delta z|$ matches the geometric prediction from the EMCal barrel radius.

3. 73% of rejected clusters are well-reconstructed photons with $E_{\mathrm{T}}/p_{\mathrm{T}}^{\mathrm{truth}} \in [0.8, 1.2]$.

4. Track ID matching alone is sufficient and definitive for truth association. The ΔR cut adds no truth information and introduces a vertex-dependent efficiency bias.

5. The $\sim$3% difference in efficiency can be assigned as a systematic uncertainty if desired, though the physics argument supports treating the track-ID-only matching as the correct baseline.

6. The $\Delta\eta$ vs. Δz correlation in full GEANT double-interaction MC (Section 3.5) confirms that the same geometric mechanism operates at larger vertex displacements (Δz RMS 48.1 cm vs. 16.9 cm), with 65% of matched clusters exceeding the $\Delta R = 0.1$ threshold.